MD. Mridul Ali Software Engineer

mridulaiub17@gmail.com +8801315238971 Bashundhara, Dhaka, Bangladesh
linkedin.com/in/md-mridul-ali- github.com/md-mridul-ali



PROFESSIONAL SUMMARY

Software Engineer with a strong foundation in software development, web application development, API development, software testing, and test automation. Skilled in C#, ASP .NET Web API, JavaScript, TypeScript, React.js, SQL, REST APIs, Playwright, Selenium WebDriver, and Postman. Experienced in developing web applications and backend APIs, designing test cases, performing functional and integration testing, and building automated end-to-end test solutions using Page Object Model (POM). Familiar with MVC, N-Tier Architecture, SDLC, STLC, Agile, CI/CD, Git, and GitHub. Pursuing software engineering opportunities to contribute to software development and quality assurance skills, building reliable, scalable, and maintainable software.

TECHNICAL SKILLS

- **Programming:** JavaScript (ES6+), TypeScript, Python, C++, C#
- **Frontend:** HTML5, CSS3, Tailwind CSS, Bootstrap, Daisy UI, React JS, Next JS
- **Software Testing:** Functional Testing, Manual Testing, Automation Testing, Unit Testing, Integration Testing, System Testing, Acceptance Testing, End-to-End Testing, Non-Functional Testing, Cross-Browser Testing
- **Test Engineering:** Test Case Design, Test Execution, Test Reports, Bug Reports, Test Planning, Page Object Model (POM), Data-Driven Testing
- **Performance Testing:** JMeter
- **Version Control & Collaboration:** Git, GitHub, GitLab, Trello, Jira
- **Linux / CLI:** Basic Linux commands, file and directory management, permissions, process management, and command-line operations
- **Backend:** ASP .NET Entity Framework, ASP .NET Web API, RESTful API Development, PHP
- **Database:** MySQL, MSSQL
- **Test Automation:** Playwright, Playwright Agents, Playwright MCP, Selenium WebDriver
- **API Testing:** Postman, REST API Testing
- **Testing Methodologies:** SDLC, STLC, Agile Methodology, Control Flow Testing, Data Flow Testing
- **CI / CD:** Familiarity with CI/CD tools such as GitLab, GitHub Actions
- **Networking:** TCP/IP, OSI Model, HTTP/HTTPS, DNS, DHCP, IP Addressing, VLSM, Common Network Protocols

PROJECTS

E-Book Management System (BookStore API)
MVC | 3-Tier Architecture | ASP .NET Web API | C# | Solo Project
- Designed and built a fully functional (CRUD, Order, and Payment) RESTful API for an e-book management system using **C# and ASP .NET Web API.**
- Implemented **MVC and N-Tier architecture** to ensure clean separation of concerns and a scalable, maintainable codebase.
- Managed relational data persistence with **MSSQL**, including schema design and query optimization.
- Used **Postman** to thoroughly test and validate all API endpoints.
- GitHub: md-mridul-ali/BookStoreUsingWebApi

E-Commerce Management System (ShopLio)
HTML | CSS | JavaScript | PHP | MySQL | Team Project
- Built a full-stack e-commerce management system using **HTML, CSS, JavaScript, PHP, and MySQL** as part of a university Web Technologies course.
- Collaborated with team members using **Git for version control**, managing tasks and merging code efficiently.
- Responsibilities included backend logic with **PHP and MySQL**, as well as front-end interface design.
- GitHub: Akib43/E-Commerce-Managment-System-Web-

English Janala – Interactive Vocabulary Learning Platform
HTML | Tailwind CSS | JavaScript | REST API | Solo Project
- Built an interactive English vocabulary learning platform where users can explore words, meanings, and synonyms.
- Used **HTML, Tailwind CSS, Daisy UI, JavaScript (ES6)**, and integrated a third-party **REST API** for word data.
- GitHub: md-mridul-ali/english-janala
- Live Demo: English Janala

Select Your Best XI
React | JavaScript | Tailwind CSS | DaisyUI
- Built a responsive cricket team selection application with **React.js**, implementing state management, props, conditional rendering, and reusable components.
- Implemented **coin-based player selection and validation** for duplicate players, insufficient balance, and maximum squad size.
- Integrated **React Toastify** for real-time notifications and designed the UI using **Tailwind CSS & DaisyUI**.
- GitHub: md-mridul-ali/select-your-best-xi
- Live Demo: select-your-best-x1

AI-Driven Test Automation — Playwright
TypeScript | Playwright MCP | LLM | GitHub Copilot | POM | Solo Project
- Built an AI-assisted end-to-end test automation framework using **Playwright, Playwright MCP, LLM, and GitHub Copilot**, generating and executing tests from structured context files and test steps
- Implemented the **Page Object Model (POM)** architecture with reusable page classes, automated test execution, failure analysis, and iterative test fixing.
- Integrated **CSV-based data-driven** testing and Playwright HTML reporting to validate multiple test datasets and review execution results.
- GitHub: md-mridul-ali/ai-driven-test-automation

Demoblaze-Ecommerce-Automation (Playwright)
Playwright | JavaScript | Git/GitHub | Solo Project
- Developed end-to-end automated test scripts for the Demoblaze e-commerce platform using **Playwright with JavaScript.**
- Automated core user workflows, including **User Registration, Login, Product Selection, Add to Cart, Checkout, and Logout** functionalities.
- Implemented **browser automation, element handling, assertions**, and test execution for functional validation.
- Created structured test scenarios to validate expected application behavior and reduce repetitive manual testing effort.
- GitHub: md-mridul-ali/demoblaze-ecommerce-automation

EDUCATION

Bachelor of Science: Computer Science and Engineering (CSE) 2026
American International University-Bangladesh (AIUB)
CGPA: 3.73

Higher Secondary Certificate: Science 2021
Rajshahi Court College
GPA: 5.00

Secondary School Certificate: Science 2019
Panka High School
GPA: 4.83

CERTIFICATES

IT Essentials — Cisco Networking Academy
Completed: December 2022

Workplace Communication Essentials — Passport to Earning Bangladesh
Completed: June 2026

SOFT SKILLS

- Problem Solving
- Time Management
- Team Collaboration
- Continuous Learning
- Communication
- Leadership

REFERENCES

DR. KHANDAKER TABIN HASAN,
Associate Dean (Faculty of Science & Technology) & Head, Computer Science (Graduate Program),
American International University-Bangladesh (AIUB)
tabin@aiub.edu, +8801721727528

MD. Mortuja Morshed, *Software Quality Assurance Engineer (SQA),* Pubali Bank PIC, Head Office
mortuja.ict@pubalibankbd.com, +8801784866522